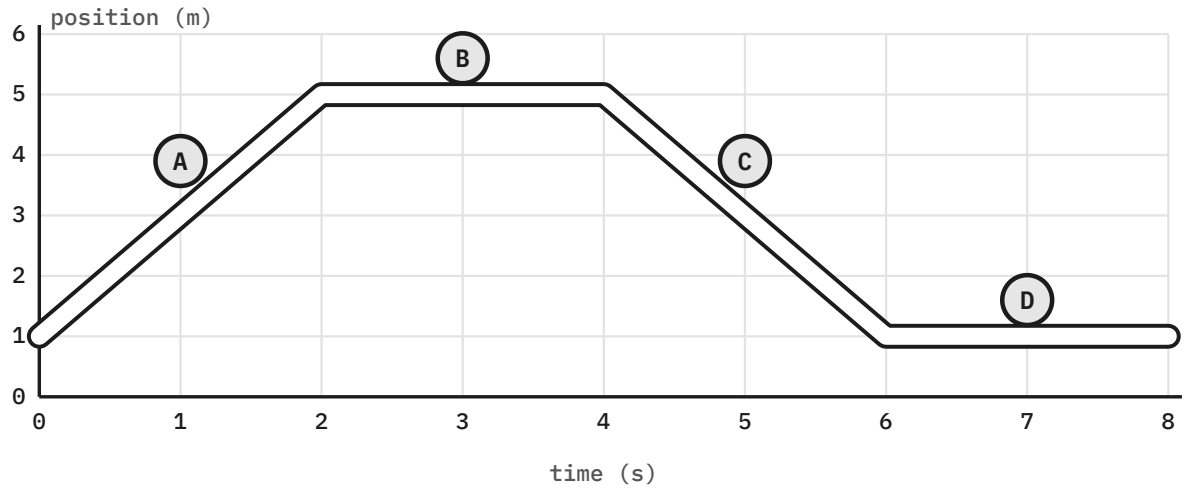
Read every part of the x–t graph

03 F

DIRECTIONS · Match each segment to its card by letter, then trace it purple: its slope is the velocity. Height = position. Slope = velocity.



A · 0–2 s
v = ____ m/s
away / back / stopped

B · 2–4 s
v = ____ m/s
away / back / stopped

C · 4–6 s
v = ____ m/s
away / back / stopped

D · 6–8 s
v = ____ m/s
away / back / stopped

SPEED OR VELOCITY? · WHOLE TRIP 0–8 s

distance ____ m · displacement ____ m
average speed = ____ m/s · average velocity = ____ m/s

TELL THE MOTION STORY

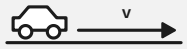
First the walker..., then..., finally...

S1 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Picture dictionary: velocity

Language goal: I can **explain** what slope means.



+5 m/s →

velocity v

velocidad · velocidade

how fast + which way

my language: _____

use it: _____



5 m/s · no direction

5 m/s

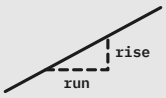
speed

rapidez · rapidez

how fast only

my language: _____

use it: _____



rise ÷ run

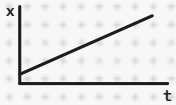
slope

pendiente · inclinação

how steep a line is

my language: _____

use it: _____



x ↑
→ t

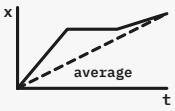
x–t graph

gráfica x–t · gráfico x–t

position on the up axis

my language: _____

use it: _____



total ÷ time

average

promedio · média

one value for a whole trip

my language: _____

use it: _____



v = 0

v = 0

at rest

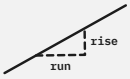
en reposo · em repouso

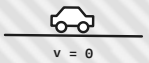
not moving

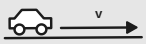
my language: _____

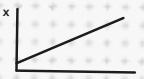
use it: _____

MATCH · WRITE THE WORD FOR EACH PICTURE









I can point and say all 6 words to a partner □

S1 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Hallway sprint – the replay

● LIVE

You're the broadcast analyst. Practice data, not a real race.

03 X

SPLIT DATA

t (s)	x (m)	Δx (m)	v (m/s)
0	0	–	–
1	2		
2	6		
3	10		
4	14		
5	16		

SLOW-MO x–t GRAPH

STAT 1 · FASTEST SECOND

Which? ____

How does the graph show it?

STAT 2 · AVERAGE v, 0–5 s

v = ____ m/s

STAT 3 · STEADY STRETCH

When was v constant? ____

COMMENTARY BOOTH

"Off the line she covers ____ m in the first second… by second 3 she's holding a steady ____ m/s – you can see it in the ____ of the line!"

S1 1 2 3

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☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Four steps, every problem

03 M

DIRECTIONS · Work each problem through all four boxes, in order.

PROBLEM 1 · A walker moves +12 m in 4 s. What is the average velocity?

1 READ

Δx = ____ · Δt = ____ · find ____

2 DRAW

x–t sketch

3 SOLVE

$v = \Delta x \div \Delta t$

4 WRITE

The average velocity is ____ m/s, meaning ____ .

PROBLEM 2 · A student walks 8 m out and 8 m back in 10 s. Find average speed and average velocity.

1 READ

distance ____ · displacement ____ · time ____

2 DRAW

route sketch

3 SOLVE

speed = distance ÷ t · $v = \Delta x \div t$

4 WRITE

Speed is ____ but velocity is ____ because ____ .

Hardest box: □ Read □ Draw □ Solve □ Write

S1 1 2 3

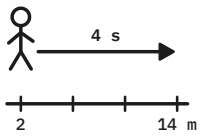
UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Talk it through: velocity

Language goal: I can **explain** a graph's slope.

DIRECTIONS · Partner A and B read their lines aloud. Listen, then write. Switch each round.

ROUND 1 · A WALKER: 2 m → 14 m IN 4 s

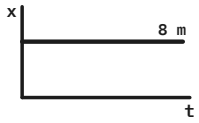


A ASKS "What is the walker's velocity?"

B SAYS "The change in position is ____ m. I divide by ____ s. So v = ____ m/s."

A CHECKS "So you're saying the walker moves ____ m every second. I agree / disagree because ____."

ROUND 2 · SWITCH · A FLAT x–t LINE AT 8 m

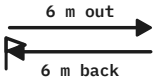


B ASKS "Is the walker fast because the line is high?"

A SAYS "No. The height shows ____. The slope is ____, so the walker is ____."

B ADDS "I want to add that velocity comes from the ____ of the line."

ROUND 3 · BOTH · A ROUND TRIP, 6 m OUT AND BACK IN 4 s



A "The average speed is ____ because the distance is ____."

B "But the average velocity is ____ because the displacement is ____."

LISTENING LOG · MY PARTNER SAID…

NOW I WRITE IT MYSELF

Velocity is different from speed because…

Stems: I agree because… · Can you say that again? · What does ___ mean? · I want to add…

S1 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Unlock the code: velocity

03 C

DIRECTIONS · Solve each station. Copy the letter beside your answer into the lock.

STATION 1

x goes from 2 m to 14 m in 4 s. Find v.

(S) +3 m/s (K) +3.5 m/s (V) 12 m/s

STATION 2

x goes from 9 m to 3 m in 2 s. Find v.

(A) +3 m/s (L) −3 m/s (U) −6 m/s

STATION 3

An x–t line is flat at x = 8 m. Find v.

(O) 0 m/s (P) 8 m/s

STATION 4

6 m east, then 6 m west, in 4 s. Average speed?

(P) 3 m/s (N) 0 m/s

STATION 5

Same trip. Average velocity?

(E) 0 m/s (R) 3 m/s

STATION 6

8,500 m/s in km/h (× 3.6)?

(S) 30,600 km/h (T) 2,361 km/h

LOCK [1] [2] [3] [4] [5] [6]

Hint: what we call the steepness of x–t graphs (plural).

S1 (1) (2) (3)

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else